

## NodeJS / Asynchronous – Exercise 7

Create an Express app with an endpoint to convert between different currencies.

The server will provide the option to run multiple "queries". Each query includes:

- **from** – string (must be a valid currency code)
- **to** – string (must be a valid currency code)
- **amount** – number

The request format is an array of queries.

The response format is an array of results, in the following format:

- **status** – boolean
- **values** – number or string: in case of success, the result number. In case of an error, the error message.

### Requirements:

- Create a models file, including the **query** and the **result** interfaces.
- Create a service layer, including two functions:
  - **convert(from, to, amount)** The function makes an API call to an external currency server to retrieve the current exchange rate between the requested currencies. Based on the response, it calculates and returns the result.
  - **runConvertQueries(queries)** – The function calls the **convert** function for each query. The conversions **should run** in parallel – not one by one. A failed query should not cause other queries to fail. (hint: `Promise.all()` )
- Create the **API layer**, including the following endpoint:
  - HTTP method - **POST**
  - URL - **/api/currencies/convert**
  - Body parameters: **Array<Query>**
  - Returns: **Array<Result>**

### Notes

- Run the app, test the endpoint from Postman – positive and negative flows, and submit the Postman collection file with the project.
- Make sure to write clean code that follows all conventions.